Portfolio Optimization: A Quantitative Approach to Maximizing Return within a Risk Threshold.

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Abstract

This project demonstrates the application of linear programming to determine the optimal allocation of a five-asset portfolio consisting of AMZN, NFLX, AAPL, MSFT, and TSLA. Historical return and volatility data were used to maximize expected return while satisfying constraints on full investment, non-negativity, diversification, and a specified risk tolerance. The model concludes that the Minimum Variance Portfolio is dominated by AAPL and MSFT, and the Maximum Return Portfolio allocates all its weight to TSLA. To illustrate the trade-off between risk and return, the Efficient Frontier was constructed. A sensitivity analysis evaluates changes in risk tolerance, allocation limits, and asset removal and examines how these deviations impact the optimal solutions. This model shows that diversification improves efficiency, and the assets that have the greatest impact on achieving the highest return and lowest risk are TSLA and MSFT. The model utilizes historical data from Yahoo Finance over the past five years and simple assumptions to provide a clear framework for understanding portfolio optimization.

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1 Introduction

1.1 Context

Portfolio optimization is the practice used by investors to find the best possible balance between maximizing expected returns and minimizing risk. This is a fundamental process in investment management, and it is flexible, allowing it to be tailored to the desired financial goals and varying levels of risk tolerance. Portfolio optimization offers a structured, quantitative approach to help investors achieve their goal of finding the optimal balance of risk and return, tailored to their individual needs. The quantitative approach is used in finance to select the optimal mix of asset allocation that achieves the greatest expected return while staying within a specified risk tolerance.

1.2 Problem Statement

Consider a young investor looking to build their portfolio and build long-term wealth. This young investor is open to a higher risk tolerance for a greater expected return on investment. The five potential assets the investor is looking to invest in are: AMZN, NFLX, AAPL, MSFT, and TSLA. We let our decision x_1, x_2, x_3, x_4, and x_5 represent the percentages of the portfolio allocated to AMAZ, NFLX, AAPL, MSFT, and TSLA, respectively. Assume the investor will allocate their full budget, so the portfolio weights must sum to 1, and all weights must be non-negative because short-selling is not allowed. There is a defined upper limit for the risk tolerance (γ). This model will help avoid extreme volatility or over-concentration in a single stock. The standard linear optimization problem takes the form $max\ c^T x\ s.t\ Ax \leq b\ and\ x_i \geq 0\ for\ i = 1, \ldots, 5$, which can be converted to its dual problem in the form $min\ b\ \ y\ st.t\ A^T y \geq c\ and\ y_j \geq 0\ for\ j = 1, \ldots, 5$.

These standard linear optimization problems will be solved to find the optimal solution x^*, which represents the recommended asset allocation that provides the highest return within the specified risk tolerance. To estimate the optimal solution using appropriate risk tolerance levels, historical price data will be found from Yahoo Finance. Daily returns, expected returns, and risk scores will be calculated from the data, based on the volatility of each asset. After these values are determined, they will serve as the inputs for the linear programming model.

The purpose of this project is to determine how a young investor should allocate their investment amount among five potential assets to achieve the highest possible expected return while keeping the total risk below their tolerance.

1.3 Project Outline

In this project, there will be a:

- Literature review: Dives into existing work and background information, such as introducing key concepts and explaining existing research relating to risk, return, diversification, efficient frontier, and optimization methods.
- Methodology: Provides the data collection process, formulates the optimization model, explains the objective function and constraints, and justifies the solution techniques used.
- Results and Discussion: Provides optimization outcomes, efficient frontier analysis, sensitivity analyses, and an evaluation of the model's performance.
- Conclusions and Recommendations: Summarizes the main findings, discusses the significance of these findings in a real-world context, acknowledges the limitations of the model, and suggests future research directions.

2 Literature Review

2.1 Background Information

Portfolio comes from broad meanings, but in business and finance, it refers to any collection of assets that can be considered investments. This meaning can be applied in other senses, such as built infrastructures, R&D projects, and patent-protected assets; however, in the context of this project, the assets refer to investments. Optimization identifies the most preferred decision alternatives out of other alternatives while satisfying constraints that reflect the decision-maker's preferences using mathematical methods. According to Salo et al. (2024), in their article "Fifty Years of Portfolio Optimization," they define portfolio optimization as the mathematical methods used to help select the best portfolios of assets, accounting for the decision-maker's preferences, relevant constraints, and uncertainties.

The core concepts of portfolio optimization are based on risk, expected return, efficient frontier, and diversification. Risk is defined as the unpredictability of an investment's actual return compared to its expected return and is typically represented by the variance of returns. This is measured through the variance of returns, representing the volatility of the assets' performance. The expected return is the measured average performance of an asset over time, based on historical data, making it suitable for an optimization model. The efficient frontier represents the set of portfolios with the highest expected return values for multiple risk tolerances. Diversification is a risk management strategy that assesses and combines assets whose returns do not move in perfect correlation to reduce portfolio volatility.

2.2 Existing Work

Portfolio optimization problems can be solved using linear programming. All the core concepts mentioned above can be directly converted into the structure of a linear optimization problem of the form $max\ c^T x\ s.t\ Ax \leq b\ and\ x_i \geq 0\ for\ i = 1, \ldots, 5$, where risk, expected return, efficient frontier, and diversification are represented. The expected return of each asset can be demonstrated by the coefficients in the objective function, as shown by the "c" in $c^T x$. In

the context of maximizing return with a given risk tolerance, the investor chooses a maximum risk, γ, as a constraint, which takes the form $d_1x_1 + d_2x_2 + \ldots \leq \gamma$. The asset allocation rules and diversification/efficient frontier are displayed as constraints in $Ax \leq b$. Since there are no negative investments in this project, we have the final constraints $x_i \geq 0$. So a portfolio optimization problem would be displayed as

Objective Function (Return)	$max\ c^T x$
Constraints $Ax \leq b$ =	*Risk constraint* *Full investment in inequalities* *Diversification bounds*
Non-negativity	$x_i \geq 0$

Markowitz (1952) is often referred to as the father of Modern Portfolio Theory (MPT); his theory provides the framework that guides investors in building a portfolio, considering the expected return and risk of assets in relation to one another. MPT's foundation is built on the core concept that investors can build a portfolio by considering the correlation between assets, rather than their individual returns, to maximize expected return at a given risk threshold or minimize risk for a given expected return. His theory highlighted diversification to maximize a portfolio's expected return for a given risk tolerance or to minimize risk for a given expected return. He introduced the idea of asset allocation, which emphasizes balancing risk and return through diversification. This showed investors evaluating their risk based on how their assets move in tandem rather than individually. Later, in other work by Markowitz (1999), he discusses the early history of portfolio theory and provides context for the development of his MPT theory. He demonstrates how his theory MPT transformed investment decision-making, and he received the Nobel Memorial Prize in Economic Sciences in 1990. His ideas form the theoretical foundation for this project, as we aim to maximize returns within a specified risk threshold and allocate assets in the most effective way to achieve this goal.

In addition, Salo et al. (2024) reviewed fifty years of research in portfolio optimization. He emphasized major methodological improvements, including enhanced risk measures, multi-objective optimization frameworks, and computational tools that enable investors to analyze large datasets more easily and effectively. From this article, it can be noticed that portfolio optimization has advanced beyond classical models to incorporate practical constraints, market imperfections, and decision-maker preferences. Salo et al. (2024) highlight the role of technology in optimization, like linear and nonlinear programming. This helps the use of computational optimization in this project. Typically, portfolio models rely on quadratic programming; however, linear programming with risk constraints is commonly used when investors want simpler, more interpretable models. Common studies with constraints such as no

negative investments, full budget allocation, and asset diversification closely align with the model constructed in this project.

2.3 Gaps

Classical mean-variance optimization is commonly used for building investment portfolios, which require quadratic programming techniques. Quadratic programming techniques are typically more complex than necessary for the basic optimization problems in this project; linear programming is used instead. Most portfolio models are designed for experienced or advanced investors. This project relates to a young investor seeking practical guidance in maximizing return. This project builds upon existing work, such as Markowitz (1952) and Salo et al. (2024), by applying linear programming and real data to create an optimal portfolio for a young investor.

3 Methodology

The methodology section outlines the mathematical and computational procedures used to solve the portfolio optimization model. The goal of this model is to optimize the allocation of an investment budget to five assets: AMZN, NFLX, AAPL, MSFT, and TLSA, that maximize the expected return while remaining in a given risk margin.

3.1 Data Collection and Processing

The data used in this project was collected from Yahoo Finance from December 2020 through December 2025. This dataset includes the daily adjusted closing prices ($P_{i,t}$) over the time period (t) for each asset (i), which can be used to calculate the daily return ($r_{i,t}$) using the standard formula:

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

where $P_{i,t}$ represents the closing price on day t, and this formula calculates the day-to-day percent change in price using the daily closing prices for each asset (i).The average expected return for each asset was computed as the sample mean, using the daily return, in the formula:

$$\mu_i = \frac{1}{T}\sum_{t=1}^{T} r_{i,t}$$

where T is the total number of observations over the time period. Lastly, the risk for each asset can be calculated using the standard deviation of the returns, computed using:

$$\sigma_i = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(r_{i,t} - \mu_i)^2}$$

The volatility values (σ_i) represent an approximation of the risk contribution of each asset in the model. The average expected return for each asset will be used in the objective function as

the coefficient to x which is the fraction of investment allocation to each asset represented by the return vector $c = (\mu_1, \mu_2, \mu_3, \mu_4, \mu_5)^T$. The risk vector $d = (\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5)^T$ will be used to formulate the risk tolerance constraint, which, together with the return and risk vectors, provides the core numerical inputs for the linear optimization model.

3.2 Model Formulation

The portfolio optimization problem is based on the Modern Portfolio Theory, which is used to determine the optimal allocation of a total investment across each asset in the portfolio.

3.2.1 Objective Function

The objective function is to maximize the expected return:

$$max\ c^T x = \mu_1 x_1 + \mu_2 x_2 + \mu_3 x_3 + \mu_4 x_4 + \mu_5 x_5$$

where $x = (x_1, x_2, x_3, x_4, x_5)^T$ represent the fraction of the total investment allocated to each of the assets: AMZN, NFLX, AAPL, MSFT, and TSLA, and the return vector (c) is the coefficients for (x).

3.2.2 Constraints

The budget constraint consists of the sum of the weights being equal to 1. This logically means that no more than 100% of the investing budget can be allocated to the assets, preventing the model from overallocating money into the assets.

$$x_1 + x_2 + x_3 + x_4 + x_5 = 1$$

The risk tolerance constraint represents the total risk of the portfolio. The risk is restricted by the maximum value (γ).

$$d^T x = \sigma_1 x_1 + \sigma_2 x_2 + \sigma_3 x_3 + \sigma_4 x_4 + \sigma_5 x_5 \leq \gamma$$

The constraint that prevents short selling, a strategy where a trader can profit if a stock's price goes down instead of up, is represented by:

$$x_i \geq 0\ for\ i = 1, \ldots, 5$$

and the constraint that creates a boundary on the maximum allocation to any given asset can be formulated as:

$$x_i \leq B$$

3.2.3 Standard Linear Programming Form

The standard linear form of the portfolio optimization model becomes:

$$max\ c^T x$$
$$s.t.\ 1^T x = 1$$

$$d^T x \leq \gamma,$$
$$0 \leq x_i \leq B, \; for\, i = 1, \ldots, 5$$

to maximize the return on a full investment in five assets: AMZN, NFLX, AAPL, MSFT, and TSLA, under different risk tolerances and constraints on weight allocation.

3.3 Solution technique

This section describes the computational framework used to solve the portfolio optimization problem. The maximum profit is calculated, along with the allocation of the total investment to each asset. The problem was solved using R, specifically the solve_gamma() function that solves the standard mean-variance portfolio optimization problem. Inside the solve_gamma() function, Dmat = g * Sigma_pd is used for the quadratic term of the optimization, and dev = -mu is used for the linear term.

3.3.1 Portfolio Optimization Under Varying Risk Tolerance Levels

To examine how the optimal solution is influenced by the risk preference of the investor, the model is solved for different risk tolerance levels represented by gamma (γ):

$$\gamma = (1,\ 10,\ 50,\ 90)$$

where $\gamma = 1$ represents the highest risk of investment, and $\gamma = 90$ represents the lowest, most conservative investment options. The risk constraint

$$d^T x \leq \gamma$$

is updated to examine the change, while the other components of the model remain the same, and there is no maximum weight constraint on how much of the total investment can be allocated to a given asset. The standard linear formulation of this example when $\gamma = 1$ would be:

$$max\ c^T x$$
$$s.t.\ 1^T x = 1$$
$$d^T x \leq 1,$$
$$x_i \geq 0, \; for\, i = 1, \ldots, 5$$

The outcomes of this model are used to understand how varying risk thresholds affect the distribution across the five assets and how influential these distribution changes are to the total return outcome.

3.3.2 Portfolio Optimization Under Varying Allocation Boundaries

The second sensitivity experiment is conducted to examine how the portfolio weights change when the maximum allocation per asset is limited to different degrees. In this experiment, the risk (γ) is set at a constant value of 10 to fully isolate the changing allocation maximum. This model is solved for the different allocation maximums (B) of

$$B = (0.3,\ 0.6,\ 0.9)$$

representing the maximum allocation per data set of 30%, 60%, and 90%, meaning that the model can not allocate more than 30%, 60%, or 90% per asset in the given model. The standard linear formation of this model when $B = 0.3$ and $\gamma = 10$ is represented by:

$$\begin{aligned} max\ & c^T x \\ s.t.\ & 1^T x = 1 \\ & d^T x \leq 10, \\ & 0 \leq x_i \leq 0.3,\ for\ i = 1, \ldots, 5 \end{aligned}$$

The outcomes of this model will illustrate how a smaller allocation maximum forces the model to distribute the investment more evenly among the assets, and how this distribution changes the total return on investment.

3.4 Methodology Justification

The methodology for this project is based on the Modern Portfolio Theory and optimization. The steps used in this optimization problem follow the basic outline for data collection, statistical estimation, and numerical analysis of the theory, while adding in constraints and boundaries that reflect real-life circumstances. The resulting portfolio allocations can then be made to accurately reflect decisions that can be made in the market.

3.4.1 Data Collection and Processing

The daily data collected for this problem were obtained from Yahoo Finance, which provides real-time financial data from stocks worldwide. Using data for different assets from the same location guarantees stability across units and time steps, allowing for the expected daily return for each asset to be consistent, in terms of accuracy and units, with the other assets.

3.4.2 Model Formulation

The Modern Portfolio Theory provides that mathematical framework for this optimization problem, where the objective function aligns with the goal of maximizing the total return on investment. A full investment is assumed, and this is consistent with the first constraint in the model:

$$s.t.\ 1^T x = 1$$

which ensures that the total allocation to all five datasets is 100% of the investment.
Non-negative and maximum allocation boundary constraints were also added to the model:

$$0 \leq x_i \leq B,\ for\ i = 1, \ldots, 5$$

representing the model's inability to short-sell, but allowing the model to still allocate 0% of the investment into a given asset. The maximum allocation boundary is used to limit the allocation to any given asset, preventing the total investment from being placed in a single asset. This implements a practical reflection into real-world portfolio management while maintaining diversification within the portfolio. The final constraint added to this optimization problem is the risk tolerance for each asset's investment.

$$d^T x \leq \gamma,$$

The risk constraint ensures that the portfolio allocation does not exceed a predetermined risk level, which can be specified by the investor. This aligns with real-world circumstances of investors wanting to maximize their outcome while not exceeding a risk threshold.

4 Results and Discussion

4.1 Descriptive Statistics

Daily adjusted closing prices for AMZN, NFLX, AAPL, MSFT, and TSLA were collected between December 1st, 2020, to December 1st, 2025, from Yahoo Finance using the *quantmod* package in the R programming language. Daily discrete returns were calculated, and the mean vector and covariance matrix were made from these series.

The following statistics show that all five assets delivered a positive average daily return over the sample period. AAPL, MSFT, and AMZN have more moderate daily return values with relatively lower volatility, while NFLX and TSLA have a higher average return but with more variance.

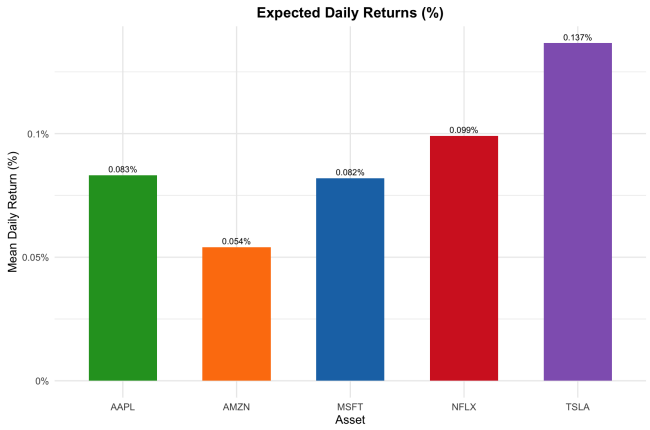


Figure 1: Expected Daily Returns %

Figure 1 shows that each asset has a positive average daily return. TSLA (0.137%) has the highest return, while AMZN (0.054%) has the lowest. This is most likely due to Tesla's rapid growth and large daily fluctuations, while Amazon is a steady, mega-cap stock with more predictable performance.

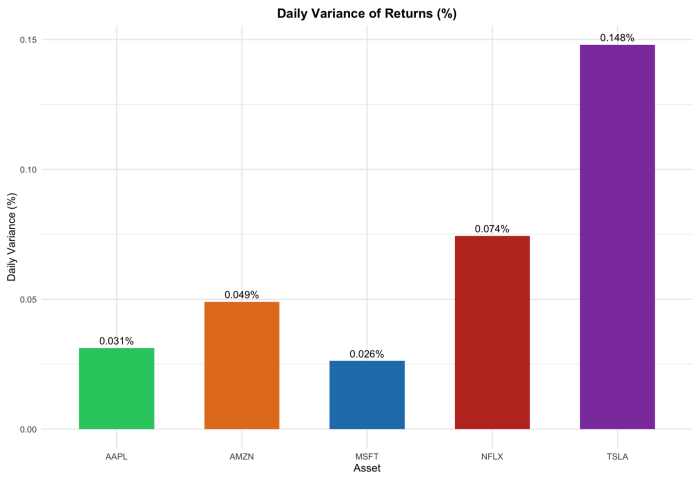


Figure 2: Daily Variance of Asset Returns

The dispersion of returns is further described in the variance measurements in Figure 2. TSLA clearly has the highest daily variance (0.148%), followed by NFLX (0.74%). In contrast, AMZN (0.031%) and MSFT (0.026%) are the least volatile. AMZN and MSFT, having the lowest volatility of the group, reflect their roles as more mature technology firms, while TSLA and NFLX are newer, higher-risk companies.

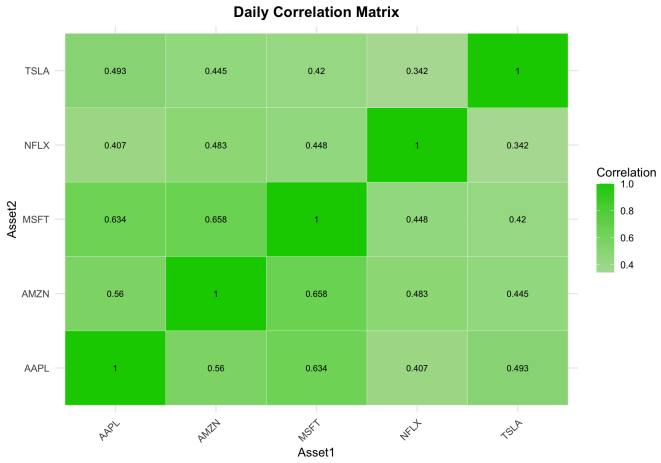


Figure 3: Daily Correlation Matrix

The dependence structure can be summarized by the correlation matrices. A positive value in Figure 3 indicates that the assets tend to move together. The higher the number, the more correlated the two assets are; a lower number indicates that they move together, but not perfectly. This imperfect movement is what ultimately allows for a diverse and lower-risk portfolio.

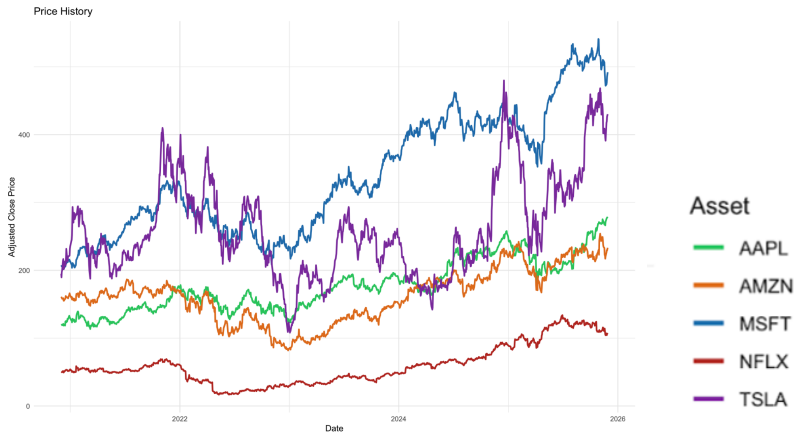


Figure 4: Price History

Finally, the raw price change over time illustrates the overall growth and volatility patterns of the different assets. Figure 4 shows clear growth across all five assets, but the magnitude and stability of that growth are not the same. MSFT has the strongest and most consistent upward trend, steadily rising over the entire period with moderate volatility. AAPL also trends upward, but with more visible fluctuations. AMZN remained relatively stable through mid-2023 before experiencing an acceleration afterward. TSLA shows significantly larger swings reflecting its high volatility. NFLX experiences its most prolonged downturn around 2022, before reversing and trending upward again. Overall, the chart shows how each stock follows a different risk–return path.

4.2 Optimization Results

Optimization was carried out in order to find portfolios that factor both risk and reward. Two main portfolios will be explored in this section: The Minimum Variance Portfolio (MVP) and the Maximum Return Portfolio (MRP). MVP looks to have a low risk, moderate reward solution, while MRP has the goal to make the most reward, not factoring in risk factors.

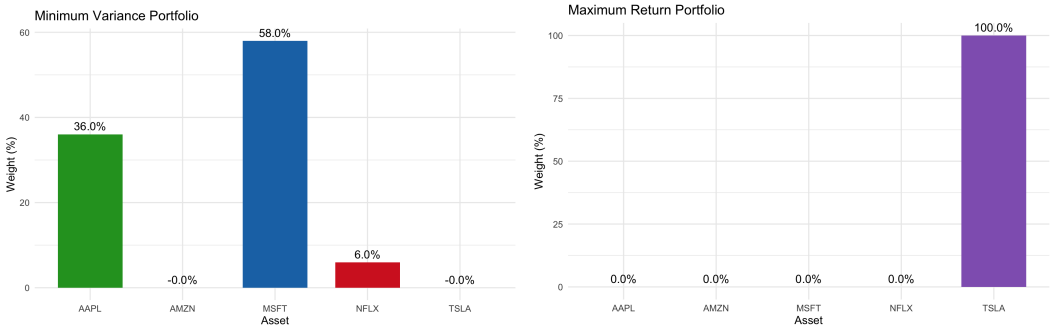


Figure 5: Minimum Variance Portfolio and Maximum Return Portfolio

The Minimum Variance Portfolio (MVP) allocates the majority of its weight to AAPL and MSFT, with a small allocation to NFLX. This is illustrated in *Figure 5*. This mix produces the lowest attainable portfolio standard deviation on the set. The Maximum Variance Portfolio

(MRP) invests 100% into TSLA, as can be seen in *Figure 6*. This is because TSLA is the asset with the highest mean return in the sample. This portfolio only looks into return and not risk, and, therefore, leans towards the assets with the highest return.

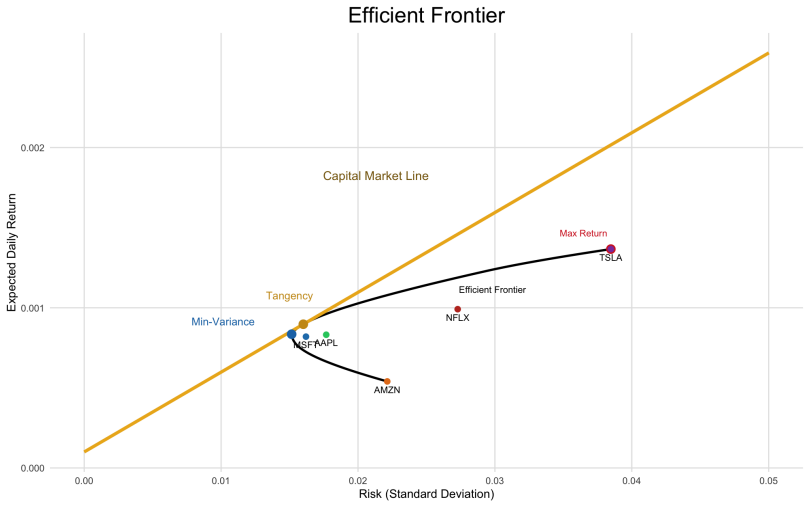


Figure 6: Efficient Frontier

The Efficient Frontier graph shows the balance between risk and return for all portfolios constructed from the assets. Each individual asset point shows its own expected daily return and volatility. TSLA has the highest return but also the greatest risk, while MSFT and AAPL are more stable and moderate performers. The black curve represents the Efficient Frontier. The Efficient Frontier is the set of optimal portfolios that achieve the maximum expected return for each level of risk. This demonstrates how having a more diverse portfolio improves the efficiency relative to just having a single asset.

The blue, leftmost point on this curve is the MVP. This approach achieves the lowest possible risk by giving heavier weighting to more stable assets. The gold tangency point marks the intersection of the Efficient Frontier and the Capital Market Line (CML). This shows the best risk-adjusted performance. The red, rightmost point indicates the MRP, which, as shown in Figure 7, is located in the same position as the TSLA point. This aligns with the results shown in *Figure 6.* Overall, the graph illustrates that diversifying a portfolio along the Efficient Frontier is more effective than relying solely on single-stock performance.

4.3 Interpretation

The Efficient Frontier in *Figure 7* shows how investors must balance risk and return. Portfolios on the left side of that black line take on very little risk, but they only offer small expected returns. As we move to the right along the frontier, higher returns become possible, but only by accepting much more volatility. This is why the MVP earns a lower return, while portfolios heavily invested in TSLA, such as MRP, earn higher returns but come with a significantly greater risk.

In the optimized portfolios, AAPL and MSFT receive the largest weights when the investor prefers lower risk. These two stocks have strong average returns, combined with relatively low volatility, and they complement other assets well because their movements are neither extremely volatile nor highly correlated. On the other hand, TSLA and NFLX only become important in higher-risk portfolios. Their returns are higher, but their price swings are much larger, so they only make sense when the investor is willing to tolerate more uncertainty to reach a higher target return.

These results also highlight the importance of diversification. Although some individual stocks, such as Tesla, are highly volatile on their own, combining them with more stable assets can help smooth out overall portfolio risk. This happens because the assets do not all move in the same way at all times. When one stock drops, another may rise or remain stable, helping reduce the total volatility of the portfolio. Diversification allows the portfolio to achieve a better risk–return trade-off than any single stock could provide on its own.

4.4 Sensitivity Analysis

In order to test the optimization's strength, several sensitivity exercises were carried out.

4.4.1 Changing the Risk-Aversion Parameter γ (Gamma)

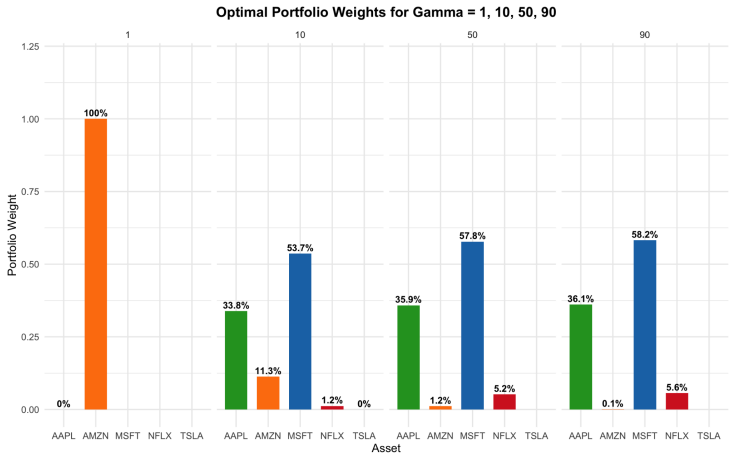


Figure 7:Optimal Weights for γ = 1,10,50,90

Portfolios were computed for γ= 1,10,50,90. As γ increases, the importance of risk becomes stronger, and the optimizer shifts from higher risk assets towards safer ones. For a low γ, the portfolio will be more aggressive, but for a higher γ, it becomes more attentive to having a low-risk portfolio.

For γ = 1, the model puts very little emphasis on risk, but it does not ignore it completely. Although TSLA has the highest return, it also exhibits the highest volatility, which renders it inefficient even when a small risk penalty is applied. AMZN, on the other hand, has a moderate return paired with much lower variance, giving it the best return-to-risk ratio in the group.

Because of this, the optimizer selects AMZN as the single most efficient asset at $\gamma = 1$, resulting in a 100% allocation.

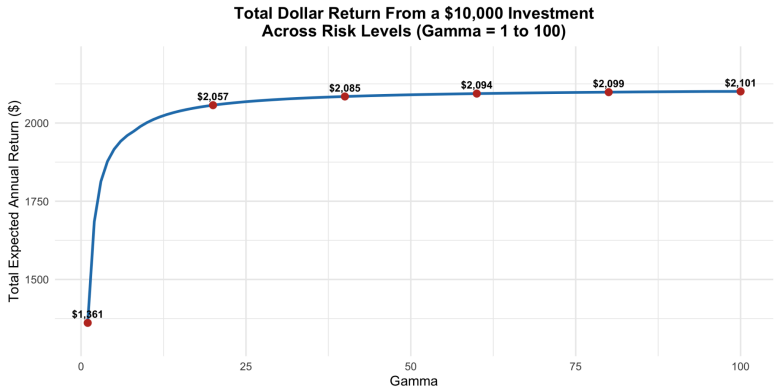


Figure 8: Total Dollar Return Across Different Gammas

The above figure shows a theoretical situation in which $10,000 is invested between the assets and what their return would be over the last 5 years as gamma is increased from 1 to 100. The biggest changes happen at the lower gamma values. This steep initial jump happens because, at a low risk aversion, the optimizer leans heavily on volatile assets. Even though these assets have high returns, their risk is large, and the portfolio becomes less efficient. An investor could see that the "optimal zone" is reached relatively quickly, around 15-20 gamma. Past this point, increasing risk aversion does not drastically hurt the total dollar return.

4.4.2 Weight Constraints (Max 30%, 60%, 90%)

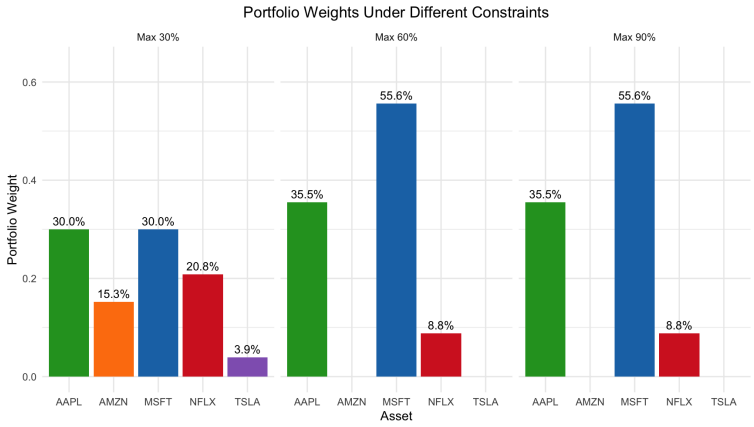


Figure 9: Portfolio Weights Under Different Constraints

This second experiment introduced upper bounds on individual asset weights. Imposing a 30% cap yields the most diversified allocation, forcing the optimizer to spread capital across multiple assets rather than concentrating it in a single stock. Relaxing the cap to 60% and 90% gradually allows more concentration, and the solutions approach the unconstrained case as the cap increases.

4.4.3 Removing an Asset

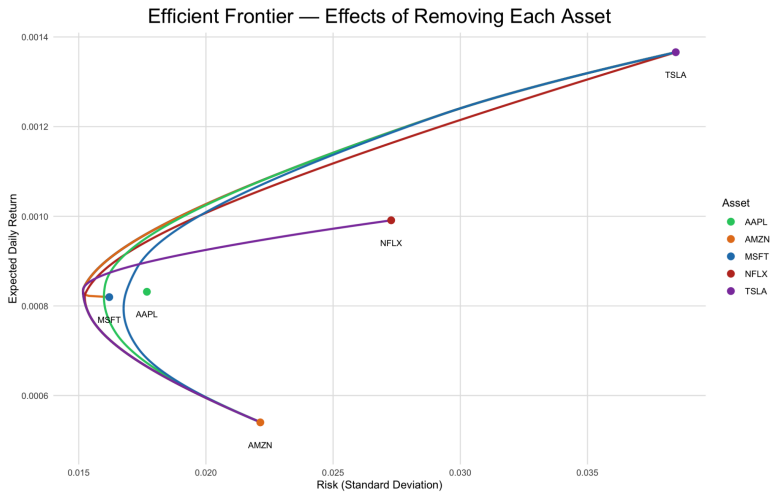


Figure 10: Efficient Frontier- Effects of Removing Each Asset

. In this graph, the frontier line moves depending on which asset was removed. The most dramatic deformation occurs when TSLA is removed. This is because the right endpoint, or the highest risk point, for each line is always the highest reward stock. In all other cases, TSLA is the highest. When TSLA is removed, the entire right side of the curve flattens and drops. This shows that, without TSLA, the portfolio loses access to the highest achievable returns. Removing MSFT produces the opposite effect. When removed, the lower-left portion of the frontier lifts upward. This reveals that MSFT was anchoring the minimum-risk region, and without it the portfolio cannot achieve the same level of stability. The curves for excluding AMZN, AAPL, and NFLX cluster tightly around the full frontier, but each diverges subtly. Taken together, it can be seen that the right side of the frontier is most affected by losing TSLA, while the left side is most affected by losing MSFT.

4.5 Evaluation

Overall, the optimization model worked well and produced results that made logical sense given the initial data. The efficient frontiers were smooth, stable, and behaved exactly as expected. The full frontier always appeared as the “best” curve, and every reduced-asset frontier fit neatly inside it. This shows that the model was computing risk and return correctly and that the solver handled the covariance matrix without numerical issues.

The sensitivity tests were especially useful in revealing how the portfolio reacts to changes. When certain assets were removed, the frontier shifted in distinct ways that matched what we saw in the actual data. For example, removing TSLA caused the high-return part of the frontier to drop sharply, while removing MSFT raised the minimum-variance end. These behaviors were clear, consistent, and easy to interpret, which suggests that the model wasn’t just producing generic theoretical shapes; it was responding accurately to the real characteristics of

each stock. Meanwhile, removing AMZN, AAPL, or NFLX only changed the frontier slightly, showing that the optimizer remained stable and was not overly sensitive to minor adjustments.

Even though the model performs well, there are limitations. For example, it relies entirely on historical data, which may not be accurate in predicting future market conditions. Despite these limitations, the model provides a clear and reliable picture of how the assets interact, how diversification works in this dataset, and how the portfolio's risk–return trade-off changes under different conditions.

5 Conclusion and Recommendations

5.1 Summary of Findings

The Minimum Variance Portfolio (MVP) consisted mostly of Apple and Microsoft, with a small allocation to Netflix. These assets consistently offered the lowest risk and strong, stable returns. This makes this a well-balanced portfolio with a low amount of risk.

In contrast, the Maximum Return Portfolio (MRP) allocates 100% to Tesla, reflecting its position as the highest-return asset in the dataset. Since the MRP ignores risk, the optimizer concentrated fully on TSLA despite its high volatility.

The Efficient Frontier showed a smooth trade-off between risk and return. The left side of the frontier (low risk) was shaped by AAPL and MSFT, while the right side (high return) was driven almost entirely by TSLA. Diversification improved portfolio efficiency, allowing higher returns at lower risk than any individual asset could achieve alone.

Sensitivity analyses revealed clear patterns. As gamma increased, allocations quickly shifted toward safer assets and stabilized around a consistent mix. Weight caps forced greater diversification at the expense of slightly lower returns, and removing assets produced predictable changes: removing TSLA reduced the maximum attainable return, while removing MSFT weakened the minimum-variance region. Removing AMZN, AAPL, or NFLX produced only small shifts, confirming overall model stability.

5.2 Real-world Implications

The portfolio outlined in this project can be applied to real-world financial decision-making problems. Although this exact project can only be used to allocate resources between the five assets selected, the framework of this project can be applied to understand assets and implement different strategies to inform investment decisions. Linear programming with risk constraint problems is commonly used as a simpler, more interpretable model by investors.

Investors face the challenge of balancing returns and potential risks when assessing the right investment options. A model that provides information on how increasing risk can enhance

returns will help inexperienced investors make informed decisions on the market. Having the ability to adjust the desired risk margins and the maximum allocation to each asset in the model enables personalization of the model to any situation, allowing the most informed decisions to be made based solely on the data.

5.3 Limitations

This project relies on historical data for the daily average return and the expected daily return. The use of past data is a limitation in itself because it may be outdated, incomplete, or inaccurate. This project can offer a simple, easily understood model for investors via linear programming, but could does not accurately present a model that would be used by companies. The data does not reflect future market behavior, especially for volatile assets. Several simplifying assumptions were made to align the model with key concepts covered throughout the semester and match the form $max\ c^T x\ s.t\ Ax \leq b\ and\ x_i \geq 0$. These assumptions limit the model's ability to represent a fully realistic portfolio optimization problem.

5.4 Future Work

Suggestions for further research in this area can enhance the applicability of this model to real-world investment scenarios. This project assumed trading is free, but this is not true in practice. To improve, the model could include costs and taxes such as fixed transaction fees, realistic penalties, and trading commissions. Another extension could be to introduce a minimum or maximum holding constraint, similar to many institutions that apply rules such as capping individual assets/minimum investments in individual assets. These additions would simulate more practical real-world scenarios. In alignment with the lectures, this project does not include short selling. However, short selling is a common practice used in real-world investing. In future work, additional constraints and restrictions could be implemented to better mimic real-world scenarios. Machine learning methods for forecasting returns or estimating covariances are another research direction. Implementing these predictions would replace historical averages in the optimization and could be used to inform real-world performance. There are many options for future research that can improve this topic.

6 References

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7 Appendices

Appendix A - Photo of data

prices_clean

	AMZN	NFLX	AAPL	MSFT	TSLA
2020-12-01	161.003997802734	50.4580001831055	119.462051391602	207.462066650391	194.919998168945
2020-12-02	160.176498413086	50.3380012512207	119.812507629395	206.656066894531	189.606674194336
2020-12-03	159.336502075195	49.7519989013672	119.676239013672	205.571807861328	197.793334960938
2020-12-04	158.128997802734	49.8310012817383	119.004531860352	205.686950683594	199.679992675781
2020-12-07	157.899993896484	51.5779991149902	120.464721679688	205.619766235352	213.919998168945
2020-12-08	158.864501953125	51.265998840332	121.078002929688	207.270202636719	216.626663208008
2020-12-09	155.210006713867	49.3600006103516	118.547035217285	203.230514526367	201.49333190918
2020-12-10	155.074493408203	50.109001159668	119.96826171875	202.002288818359	209.023330688477
2020-12-11	155.820999145508	50.3219985961914	119.160293579102	204.631454467773	203.330001831055
2020-12-14	157.848495483398	52.242000579834	118.547035217285	205.533416748047	213.276672363281
2020-12-15	158.255996704102	51.9780006408691	124.485084533691	205.466247558594	211.08332824707
2020-12-16	162.048004150391	52.4830017089844	124.416931152344	210.407867431641	207.589996337891
2020-12-17	161.804000854492	53.2900009155273	125.283309936523	210.542205810547	218.633331298828
2020-12-18	160.082504272461	53.4449996948242	123.297477722168	209.745803833008	231.66667175293
2020-12-21	160.309005737305	52.890998840332	124.825790405273	213.583999633789	216.619995117188
2020-12-22	160.32600402832	52.7330017089844	128.378890991211	214.879333496094	213.446670532227

Appendix B — Code Listings (Final Non-Duplicated Version)

B1. Libraries

```
library(quantmod)
library(tidyverse)
library(PerformanceAnalytics)
library(quadprog)
library(knitr)
library(reshape2)
```

B2. Data Import

```
tickers <- c("AMZN","NFLX","AAPL","MSFT","TSLA")
getSymbols(tickers, src="yahoo",
        from="2020-12-01", to="2025-12-01")
prices <- do.call(merge, lapply(tickers, \(x) Ad(get(x))))
colnames(prices) <- tickers
```

B3. Returns, Expected Returns, Covariance, Correlation, Variance

```
returns <- na.omit(ROC(prices, type="discrete"))
expected_returns_daily  <- colMeans(returns)
expected_returns_annual <- expected_returns_daily * 252
cov_daily  <- cov(returns)
cov_annual <- cov_daily * 252
cor_daily  <- cor(returns)
daily_variance <- apply(returns, 2, var)
mu <- expected_returns_daily
Sigma <- cov_daily
Sigma_pd <- Sigma + diag(1e-6, length(mu))
assets <- names(mu)
n_assets <- length(assets)
```

B4. Minimum-Variance and Maximum-Return Portfolios

```
### Minimum-Variance Portfolio
Dmat <- 2 * Sigma_pd
dvec <- rep(0, n_assets)
Aeq <- matrix(1, nrow=n_assets, ncol=1)
Aineq <- diag(n_assets)
Amat <- cbind(Aeq, Aineq)
bvec <- c(1, rep(0, n_assets))
minvar_sol <- solve.QP(Dmat, dvec, Amat, bvec, meq=1)
w_minvar <- minvar_sol$solution
ret_minvar <- sum(w_minvar * mu)
sd_minvar  <- sqrt(t(w_minvar) %*% Sigma %*%
w_minvar)
### Maximum-Return Portfolio
idx_max <- which.max(mu)
w_maxret <- rep(0, n_assets)
w_maxret[idx_max] <- 1
ret_maxret <- mu[idx_max]
sd_maxret  <- sqrt(Sigma[idx_max, idx_max])
```

B5. Efficient Frontier (Single, Clean Version)

```
target_returns <- seq(min(mu), max(mu), length.out=200)
ef_risk <- c()
ef_ret  <- c()
for (tr in target_returns) {
  Dmat <- 2 * Sigma_pd
  dvec <- rep(0, n_assets)
  Aeq  <- cbind(rep(1, n_assets), mu)
  Aineq <- diag(n_assets)
  Amat <- cbind(Aeq, Aineq)
  bvec <- c(1, tr, rep(0, n_assets))
  sol <- solve.QP(Dmat, dvec, Amat, bvec, meq=2)
  w <- sol$solution
  ef_risk <- c(ef_risk, sqrt(t(w) %*% Sigma %*% w))
  ef_ret  <- c(ef_ret, sum(w * mu))
}
efficient_df <- data.frame(Risk = ef_risk, Return = ef_ret)
min_idx <- which.min(efficient_df$Risk)
lower_df <- efficient_df[1:min_idx, ]
upper_df <- efficient_df[min_idx:nrow(efficient_df), ]
rf <- 0.0001
sharpe_vals <- (efficient_df$Return - rf) / efficient_df$Risk
tan_idx <- which.max(sharpe_vals)
sd_tangent  <- efficient_df$Risk[tan_idx]
ret_tangent <- efficient_df$Return[tan_idx]
max_risk <- max(efficient_df$Risk) * 1.3
slope_cml <- (ret_tangent - rf) / sd_tangent
cml_df <- data.frame(
  Risk   = seq(0, max_risk, length.out=200),
  Return = rf + slope_cml * seq(0, max_risk,
length.out=200)
)
assets_df <- data.frame(
  Asset  = assets,
  Risk   = sqrt(diag(Sigma)),
  Return = mu
)
```

B6. Gamma Optimization (Only Once)

```
solve_gamma <- function(gamma){
  Dmat <- gamma * Sigma_pd
  dvec <- -mu
  Aeq  <- matrix(1, nrow=n_assets, ncol=1)
  Aineq <- diag(n_assets)
  Amat <- cbind(Aeq, Aineq)
  bvec <- c(1, rep(0, n_assets))
  w <- solve.QP(Dmat, dvec, Amat, bvec, meq=1)$solution
  tibble(Gamma = gamma, Asset = assets, Weight = w)
}
gamma_results <- map_df(c(1, 10, 50, 90), solve_gamma)
```

B7. Weight Constraint Solver (Only Once)

```
solve_weight_limit <- function(mu, Sigma_pd, gamma,
max_weight){
  n <- length(mu)
  Dmat <- gamma * Sigma_pd
  dvec <- mu
  Aeq <- matrix(1, nrow=n, ncol=1)
  beq <- 1
  A_ge0 <- diag(n)
  b_ge0 <- rep(0, n)
  A_leMax <- -diag(n)
  b_leMax <- rep(-max_weight, n)
  Amat <- cbind(Aeq, A_ge0, A_leMax)
  bvec <- c(beq, b_ge0, b_leMax)
  sol <- solve.QP(Dmat, dvec, Amat, bvec, meq=1)
  tibble(Asset = names(mu), Weight = sol$solution)
}
w30 <- solve_weight_limit(mu, Sigma_pd, gamma=10,
max_weight=0.30)
w60 <- solve_weight_limit(mu, Sigma_pd, gamma=10,
max_weight=0.60)
w90 <- solve_weight_limit(mu, Sigma_pd, gamma=10,
max_weight=0.90)
```

B8. Remove-Asset Efficient Frontier Solver (Only Once)

```
smooth_frontier <- function(asset_subset){
  idx <- which(assets %in% asset_subset)
  mu_sub <- mu[idx]
  Sigma_sub <- Sigma_pd[idx, idx]
  n_sub <- length(idx)
  target_returns <- seq(min(mu_sub), max(mu_sub),
length.out=200)
  ef_risk <- c()
  ef_ret  <- c()
  for (tr in target_returns) {
    Dmat <- 2 * Sigma_sub
    dvec <- rep(0, n_sub)
    Aeq <- cbind(rep(1, n_sub), mu_sub)
    Aineq <- diag(n_sub)
    Amat <- cbind(Aeq, Aineq)
    bvec <- c(1, tr, rep(0, n_sub))
    sol <- solve.QP(Dmat, dvec, Amat, bvec, meq=2)
    w <- sol$solution
    ef_risk <- c(ef_risk, sqrt(t(w) %*% Sigma_sub %*% w))
    ef_ret  <- c(ef_ret, sum(w * mu_sub))
  }
  tibble(Risk = ef_risk, Return = ef_ret)
}
full_frontier   <- smooth_frontier(assets)
no_AMZN_front   <-
smooth_frontier(setdiff(assets,"AMZN"))
no_NFLX_front   <-
smooth_frontier(setdiff(assets,"NFLX"))
no_AAPL_front   <-
smooth_frontier(setdiff(assets,"AAPL"))
no_MSFT_front   <-
smooth_frontier(setdiff(assets,"MSFT"))
no_TSLA_front   <-
smooth_frontier(setdiff(assets,"TSLA")
```